

PREPARED FOR

123 INC

SAMPLE

39.21471, -122.01399

T16N, R1W, Mount Diablo Meridian

✓ CORS Elev Delta: 97 ft above nearest CORS station (P270, 3.0 mi) – Good baseline geometry.

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
39.2147°N 122.0140°W WGS84	58 ft NAVD88 · MSL	CA 2 (0402) · N 2,204,199.85 E 6,557,704.40 (US ft, NAD83)	Zone X N/A	13.1° E WMM 2026

APN 001-073-001-000 | ZONING 0-S | ACRES 1.909 ac

BEST GNSS WINDOW Friday Jul 17 · 3:00 PM – 7:00 PM | PDOP 0.91 · 34 sv | NEAREST CORS
P270 · 3.0 mi ✓

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VIEW FIELD MAP

SITE SATELLITE

EMERGENCY MAP

SCAN FOR KMZ
Site file · Google Earth
Download KMZ

ASSESSOR-RECORDER
Colusa County
Clerk-Recorder
546 Jay St
Colusa, CA 95932
NEAREST BN
KT0441 · 0 mi
NEAREST CORS
P270 · 3.0 mi
GENERATED
July 17,
2026
SYSTEM
TopoScout ·
info@toposcout.com



INTERACTIVE REPORT

This report includes clickable links and QR codes throughout – tap or scan them to open maps, official data sources, and field tools directly. Your site is also available as a drafting-ready CAD (DXF) file with georeferenced aerials. Add it anytime at toposcout.com.

SITE & SIGNAL REPORT

COORDINATES

39.2147, -122.0140

WGS84 / NAD83

ELEVATION

58 ft

Above MSL (approx)

MAGNETIC DECLINATION

13.1° E

Compass reads 13.1° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel N/A

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.1

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

25.4 TECU

+2.2 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.1). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Fri, Jul 17	0.99	3:00 PM – 7:00 PM	34	Ideal
Sat, Jul 18	0.98	1:00 PM – 5:00 PM	35	Ideal
Sun, Jul 19	0.98	3:00 PM – 7:00 PM	34	Ideal
Mon, Jul 20	0.99	3:00 PM – 7:00 PM	34	Ideal
Tue, Jul 21	0.98	3:00 PM – 7:00 PM	33	Ideal
Wed, Jul 22	1.00	4:00 PM – 8:00 PM	33	Good
Thu, Jul 23	1.00	7:00 AM – 11:00 AM	31	Good

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

Days analyzed: 7 Recommended work days: 5 of 7 Best PDOP: 0.98

►

RECOMMENDED WINDOW

Thu, Jul 23

8:00 AM – 12:00 PM

Avg wind 1 mph · Gusts 4 mph

Ideal conditions — 4-hr static occupation

ALTERNATE WINDOW

Tue, Jul 21

8:00 AM – 12:00 PM

Avg wind 1 mph · Gusts 3 mph

Ideal conditions — 4-hr static occupation

🔥

WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 17, 2026) · Conditions may change – verify before fieldwork

📶

GPS CONSTELLATION HEALTH

⚠️ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

🌿 Land Cover & Multipath Risk

SKY OBSTRUCTION

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

MULTIPATH RISK

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13°

Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–1.0 across survey windows

PDOP remains ≤ 1.0 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 95°F

Low 57°F

Wind max 14 mph

Gusts 21 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:32 PM

Waxing Crescent 8%

GNSS: Avg PDOP **0.99** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.91, 34 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	1.04		30 sv		GOOD
7:00 AM	1.12		27 sv		GOOD
8:00 AM	0.91		35 sv		IDEAL
9:00 AM	0.96		32 sv		IDEAL
10:00 AM	0.93		31 sv		IDEAL
11:00 AM	0.94		29 sv		IDEAL
12:00 PM	0.95		30 sv		IDEAL
1:00 PM	1.13		27 sv		GOOD
2:00 PM	1.03		31 sv		GOOD
3:00 PM	0.91		35 sv		IDEAL
4:00 PM	0.92		35 sv		IDEAL
5:00 PM	0.91		34 sv		IDEAL
6:00 PM	0.91		31 sv		IDEAL
7:00 PM	1.26		24 sv		GOOD



GO – GOOD CONDITIONS

Clear sky | Wind: 13.8 mph (gusts 21 mph) | Precip: 0" | Temp: 57°–94.5°F



DRONE WIND ANALYSIS



CAUTION Peak: 13.8/21 mph SSE

* Sunrise: 5:56 AM ☾ Sunset: 8:32 PM fly window clipped to daylight

✓ Best flight window: 6:00 AM – 7:00 PM (13 hrs)

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	3/5 mph		NNE
6 AM	4/7 mph		ESE
7 AM	1/2 mph		NW
8 AM	1/3 mph		NNE
9 AM	4/5 mph		NNE
10 AM	4/4 mph		NNE
11 AM	1/1 mph		NNW
12 PM	3/4 mph		SSW
1 PM	7/8 mph		S
2 PM	8/9 mph		S
3 PM	10/11 mph		S
4 PM	12/14 mph		SSE
5 PM	13/16 mph		SSE
6 PM	14/18 mph		SSE
7 PM	14/21 mph		SSE
8 PM	12/18 mph		SSE

BEST WINDOW

6:00 AM – 7:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ► FREE TO FLY

Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#2

Saturday, Jul 18

WEEKEND

Clear sky

13 mph

GO

High 95°F

Low 55°F

Wind max 13 mph

Gusts 21 mph

Precip 0.00"

Sunrise 5:56 AM

Sunset 8:31 PM

Waxing Crescent 15%

GNSS: Avg PDOP 0.98 · Best 4-hr window 1:00 PM – 5:00 PM (PDOP 0.93, 35 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	1.01	31 sv	GOOD
7:00 AM	1.02	32 sv	GOOD
8:00 AM	1.03	32 sv	GOOD
9:00 AM	0.91	33 sv	IDEAL
10:00 AM	0.98	31 sv	IDEAL
11:00 AM	1.06	28 sv	GOOD
12:00 PM	0.96	30 sv	IDEAL
1:00 PM	0.97	32 sv	IDEAL
2:00 PM	0.91	35 sv	IDEAL
3:00 PM	0.91	35 sv	IDEAL
4:00 PM	0.93	36 sv	IDEAL
5:00 PM	0.99	32 sv	IDEAL
6:00 PM	0.91	34 sv	IDEAL
7:00 PM	1.18	29 sv	GOOD



GO - GOOD CONDITIONS

Clear sky | Wind: 12.6 mph (gusts 21.3 mph) | Precip: 0" | Temp: 54.9°–94.5°F



DRONE WIND ANALYSIS



CAUTION Peak: 12.6/21.3 mph SSE

* Sunrise: 5:56 AM Sunset: 8:31 PM fly window clipped to daylight

Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	=turbulence	DIR
5 AM	3/12 mph		WSW
6 AM	2/10 mph		S
7 AM	3/8 mph		SSW
8 AM	1/2 mph		SE
9 AM	2/7 mph		NW
10 AM	2/5 mph		NNW
11 AM	0/0 mph		W
12 PM	3/4 mph		SSW
1 PM	6/6 mph		SSW
2 PM	7/7 mph		S
3 PM	8/9 mph		S
4 PM	10/11 mph		S
5 PM	11/13 mph		SSE
6 PM	11/14 mph		SSE
7 PM	11/16 mph		SSE
8 PM	11/17 mph		SSE

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ► FREE TO FLY

Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#3

Sunday, Jul 19

Overcast

13 mph

GO

WEEKEND

High 106°F

Low 63°F

Wind max 13 mph

Gusts 22 mph

Precip 0.00"

Sunrise 5:57 AM

Sunset 8:31 PM

Waxing Crescent 23%

GNSS: Avg PDOP **0.98** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.94, 34 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	0.96		34 sv	IDEAL
7:00 AM	0.93		31 sv	IDEAL
8:00 AM	0.93		33 sv	IDEAL
9:00 AM	0.94		31 sv	IDEAL
10:00 AM	1.09		28 sv	GOOD
11:00 AM	1.14		25 sv	GOOD
12:00 PM	0.99		28 sv	IDEAL
1:00 PM	0.95		31 sv	IDEAL
2:00 PM	0.99		33 sv	IDEAL
3:00 PM	0.95		33 sv	IDEAL
4:00 PM	0.95		36 sv	IDEAL
5:00 PM	0.89		35 sv	IDEAL
6:00 PM	0.95		32 sv	IDEAL
7:00 PM	1.07		29 sv	GOOD

GO — GOOD CONDITIONS

Partly cloudy | Wind: 12.5 mph (gusts 21.9 mph) | Precip: 0" | Temp: 63°–105.8°F

FIELD SAFETY WARNINGS

EXTREME HEAT 105.8°F — Heat safety protocols required

DRONE WIND ANALYSIS

CAUTION Peak: 12.5/21.9 mph SSE

* Sunrise: 5:57 AM ☾ Sunset: 8:31 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 9:00 PM (15 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	3/5 mph		ENE
6 AM	4/4 mph		N
7 AM	5/6 mph		NNW
8 AM	5/6 mph		NNW
9 AM	5/5 mph		NNW
10 AM	4/4 mph		NNW
11 AM	1/1 mph		N
12 PM	2/1 mph		NW
1 PM	5/3 mph		SSE
2 PM	5/6 mph		S
3 PM	7/8 mph		S
4 PM	6/8 mph		S
5 PM	7/7 mph		SSE
6 PM	11/7 mph		SSE
7 PM	13/10 mph		S
8 PM	8/17 mph		S

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ► FREE TO FLY
Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 105°F

Low 71°F

Wind max 16 mph

Gusts 22 mph

Precip 0.00"

Sunrise 5:58 AM

Sunset 8:30 PM

Waxing Crescent 33%

GNSS: Avg PDOP **0.99** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.93, 34 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	0.91		34 sv		IDEAL
7:00 AM	0.92		35 sv		IDEAL
8:00 AM	0.93		35 sv		IDEAL
9:00 AM	1.00		30 sv		GOOD
10:00 AM	0.98		28 sv		IDEAL
11:00 AM	1.23		24 sv		GOOD
12:00 PM	0.93		29 sv		IDEAL
1:00 PM	1.08		26 sv		GOOD
2:00 PM	1.02		32 sv		GOOD
3:00 PM	0.98		34 sv		IDEAL
4:00 PM	0.94		33 sv		IDEAL
5:00 PM	0.92		33 sv		IDEAL
6:00 PM	0.87		35 sv		IDEAL
7:00 PM	1.19		28 sv		GOOD

GO – GOOD CONDITIONS
Partly cloudy | Wind: 15.9 mph (gusts 21.9 mph) | Precip: 0" | Temp: 70.6°–105.1°F

FIELD SAFETY WARNINGS
 EXTREME HEAT 105.1°F — Heat safety protocols required

DRONE WIND ANALYSIS

CAUTION Peak: 15.9/21.9 mph S

* Sunrise: 5:58 AM ☾ Sunset: 8:30 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 5:00 PM (11 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	7/6 mph		SE
6 AM	5/6 mph		ESE
7 AM	4/3 mph		SE
8 AM	3/2 mph		ESE
9 AM	2/4 mph		ENE
10 AM	0/4 mph		N
11 AM	2/4 mph		SSE
12 PM	3/4 mph		S
1 PM	5/6 mph		S
2 PM	7/9 mph		SSE
3 PM	9/10 mph		SSE
4 PM	13/11 mph		SW
5 PM	16/13 mph		WSW
6 PM	14/15 mph		SW
7 PM	14/17 mph		SW
8 PM	9/22 mph		SSW

BEST WINDOW
6:00 AM – 5:00 PM
Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS G** ▶ **FREE TO FLY**
Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ **NO TFRs WITHIN 30 MI** [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 106°F

Low 69°F

Wind max 16 mph

Gusts 23 mph

Precip 0.00"

Sunrise 5:59 AM

Sunset 8:29 PM

Waxing Crescent 43%

GNSS: Avg PDOP **0.98** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.94, 33 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	0.91		33 sv	IDEAL
7:00 AM	0.89		36 sv	IDEAL
8:00 AM	0.93		33 sv	IDEAL
9:00 AM	1.15		29 sv	GOOD
10:00 AM	1.01		31 sv	GOOD
11:00 AM	1.07		28 sv	GOOD
12:00 PM	0.99		29 sv	IDEAL
1:00 PM	1.06		28 sv	GOOD
2:00 PM	0.98		31 sv	IDEAL
3:00 PM	0.87		35 sv	IDEAL
4:00 PM	0.94		36 sv	IDEAL
5:00 PM	1.00		31 sv	GOOD
6:00 PM	0.96		30 sv	IDEAL
7:00 PM	1.02		29 sv	GOOD

GO – GOOD CONDITIONS
Partly cloudy | Wind: 16 mph (gusts 23.3 mph) | Precip: 0" | Temp: 68.9°–106.4°F

FIELD SAFETY WARNINGS
 EXTREME HEAT 106.4°F — Heat safety protocols required

DRONE WIND ANALYSIS

CAUTION Peak: 16/23.3 mph S

* Sunrise: 5:59 AM ☾ Sunset: 8:29 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 7:00 PM (13 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	4/1 mph		ESE
6 AM	5/3 mph		E
7 AM	2/3 mph		ESE
8 AM	1/2 mph		ESE
9 AM	0/1 mph		N
10 AM	0/2 mph		W
11 AM	1/3 mph		SW
12 PM	2/4 mph		SSW
1 PM	4/7 mph		SSE
2 PM	6/8 mph		SSE
3 PM	8/10 mph		SSE
4 PM	8/11 mph		SSE
5 PM	9/13 mph		SSE
6 PM	13/14 mph		SSW
7 PM	16/13 mph		SW
8 PM	16/19 mph		WSW

BEST WINDOW
6:00 AM – 7:00 PM
Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS G** ▶ FREE TO FLY
Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ **NO TFRs WITHIN 30 MI** [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 101°F

Low 72°F

Wind max 18 mph

Gusts 20 mph

Precip 0.00"

Sunrise 6:00 AM

Sunset 8:29 PM

First Quarter 53%

GNSS: Avg PDOP **1.00** · Best 4-hr window 4:00 PM – 8:00 PM (PDOP 0.93, 33 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	1.09		30 sv	GOOD
7:00 AM	1.05		30 sv	GOOD
8:00 AM	0.88		37 sv	IDEAL
9:00 AM	1.00		31 sv	GOOD
10:00 AM	1.02		29 sv	GOOD
11:00 AM	1.17		25 sv	GOOD
12:00 PM	0.97		28 sv	IDEAL
1:00 PM	1.05		28 sv	GOOD
2:00 PM	0.98		34 sv	IDEAL
3:00 PM	1.08		33 sv	GOOD
4:00 PM	0.90		35 sv	IDEAL
5:00 PM	0.93		33 sv	IDEAL
6:00 PM	0.96		31 sv	IDEAL
7:00 PM	0.94		32 sv	IDEAL

GO — GOOD CONDITIONS
Partly cloudy | Wind: 18 mph (gusts 19.9 mph) | Precip: 0" | Temp: 72°–101°F

FIELD SAFETY WARNINGS

HIGH HEAT 101°F — Monitor crew for heat stress

DRONE WIND ANALYSIS

CAUTION Peak: 18/19.9 mph SSE

* Sunrise: 6:00 AM ☾ Sunset: 8:29 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 4:00 PM (10 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤=turbulence	DIR
5 AM	8/10 mph		SE
6 AM	7/9 mph		SE
7 AM	6/9 mph		SE
8 AM	6/9 mph		SE
9 AM	6/9 mph		SE
10 AM	7/10 mph		SSE
11 AM	8/11 mph		SSE
12 PM	11/12 mph		SSE
1 PM	14/13 mph		SSE
2 PM	15/13 mph		SSE
3 PM	15/13 mph		S
4 PM	17/13 mph		SSW
5 PM	18/14 mph		SSW
6 PM	17/16 mph		SSW
7 PM	13/18 mph		SW
8 PM	10/18 mph		SSW

BEST WINDOW
6:00 AM – 4:00 PM
Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS G** ▶ FREE TO FLY
Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ **NO TFRs WITHIN 30 MI** [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 104°F

Low 60°F

Wind max 11 mph

Gusts 20 mph

Precip 0.00"

Sunrise 6:00 AM

Sunset 8:28 PM

First Quarter 64%

GNSS: Avg PDOP **1.00** · Best 4-hr window 7:00 AM – 11:00 AM (PDOP 0.97, 31 sats avg)

GNSS / DOP - HOURLY				
	PDOP		SATS	RATING
6:00 AM	1.03		30 sv	GOOD
7:00 AM	0.93		34 sv	IDEAL
8:00 AM	0.94		32 sv	IDEAL
9:00 AM	1.08		27 sv	GOOD
10:00 AM	0.92		31 sv	IDEAL
11:00 AM	1.09		26 sv	GOOD
12:00 PM	0.98		31 sv	IDEAL
1:00 PM	0.99		30 sv	IDEAL
2:00 PM	0.97		31 sv	IDEAL
3:00 PM	1.00		33 sv	GOOD
4:00 PM	1.00		32 sv	GOOD
5:00 PM	0.98		33 sv	IDEAL
6:00 PM	0.95		31 sv	IDEAL
7:00 PM	1.10		26 sv	GOOD



GO – GOOD CONDITIONS

Clear sky | Wind: 11.1 mph (gusts 19.7 mph) | Precip: 0" | Temp: 60.3°–103.9°F



FIELD SAFETY WARNINGS

 HIGH HEAT 103.9°F — Monitor crew for heat stress



DRONE WIND ANALYSIS

☒ GO Peak: 11.1/19.7 mph SSE

* Sunrise: 6:00 AM Sunset: 8:28 PM fly window clipped to daylight

☒ Best flight window: 6:00 AM – 9:00 PM (15 hrs)

 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤ turbulence	DIR
5 AM	6/3 mph		SE
6 AM	5/3 mph		ESE
7 AM	3/3 mph		ESE
8 AM	2/4 mph		E
9 AM	1/3 mph		NE
10 AM	1/2 mph		NNW
11 AM	1/1 mph		NW
12 PM	1/2 mph		SW
1 PM	3/4 mph		S
2 PM	5/5 mph		S
3 PM	7/7 mph		SSE
4 PM	10/8 mph		SSE
5 PM	11/10 mph		SSE
6 PM	10/13 mph		S
7 PM	10/18 mph		SSW
8 PM	10/20 mph		SSW

BEST WINDOW

6:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS G ▶ FREE TO FLY

Class G uncontrolled airspace - fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

GNSS SURVEY — COLUSA COUNTY, CA

39.2147, -122.0140 | Report Date: Friday, July 17, 2026

=====

TODAY (FRIDAY, JULY 17)

=====

STATUS: GO

RATIONALE

PDOP 0.99 average, excellent geometry (best window 3:00 PM–7:00 PM at PDOP 0.91, 34 sats). Clear skies, low wind (14 mph max), no precipitation. Moderate ionospheric activity (Kp 4.7 peak) manageable for static work with standard occupation windows. PRN 26 maintenance window (00:35–12:35 UTC) does not affect afternoon operations. Deploy immediately if mobilization complete.

=====

MONDAY CALL (JULY 20)

=====

STATUS: GO

RATIONALE

Monday ranks #2 overall (see Week Ranking below). PDOP 0.99, best window 3:00 PM–7:00 PM (PDOP 0.93, 34 sats). Overcast but no precipitation; wind 16 mph (manageable). Crews should plan full mobilization today and stage equipment for Monday afternoon operations. No satellite health or weather blockers.

=====

WEEK RANKING — 7-DAY GNSS SURVEY CONDITIONS

=====

1. FRIDAY, JUL 17 — PDOP 0.99 | Clear | Kp 4.7 | GO (IDEAL — start immediately if ready)
2. MONDAY, JUL 20 — PDOP 0.99 | Overcast | Kp est. 2.0 | GO (primary work day)
3. TUESDAY, JUL 21 — PDOP 0.98 | Overcast | Kp est. 2.0 | GO (secondary work day)
4. SATURDAY, JUL 18 [WEEKEND] — PDOP 0.98 | Clear | Kp est. 2.5 | CAUTION (weekend; good geometry/weather if needed)
5. SUNDAY, JUL 19 [WEEKEND] — PDOP 0.98 | Overcast | Kp est. 2.2 | CAUTION (weekend; acceptable geometry, clouds minor impact)
6. WEDNESDAY, JUL 22 — PDOP 1.00 | Overcast | Kp est. 2.0 | CAUTION (slightly elevated PDOP; wind 18 mph)
7. THURSDAY, JUL 23 — PDOP 1.00 | Clear | Kp est. 2.0 | CAUTION (morning-only optimal window, worst PDOP, 31 sats)

TOP 5 RECOMMENDED WORK WINDOWS: Friday, Monday, Tuesday, Saturday (if weekend crew available), Sunday (if needed).

=====

BEST STATIC WINDOW — FULL WEEK

=====

MONDAY, JULY 20 | 3:00 PM – 7:00 PM (4 consecutive hours)

PDOP: 0.93 (stable throughout), 34 satellites, overcast (no direct sun stress), wind 16 mph manageable. Ionospheric conditions improved vs. Friday (Kp ~2.0 est.). Recommend this as primary static baseline/control point session.

ALTERNATE: Friday, July 17, 3:00 PM–7:00 PM (PDOP 0.91, clearer skies, but Kp 4.7 peak — acceptable, requires longer occupation time per ionospheric memo).

=====

DRONE ASSESSMENT — DAY-BY-DAY

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FRIDAY, JUL 17 — FLYABLE | Clear, 14 mph wind, excellent visibility

SATURDAY, JUL 18 — FLYABLE | Clear, 13 mph wind, excellent visibility [WEEKEND]

SUNDAY, JUL 19 — CAUTION | Overcast, 13 mph wind, reduced visibility ceiling [WEEKEND]

MONDAY, JUL 20 — CAUTION | Overcast, 16 mph wind, limited sight-line clarity

TUESDAY, JUL 21 — CAUTION | Overcast, 16 mph wind, limited ceiling/visibility

WEDNESDAY, JUL 22 — NO-FLY | Overcast, 18 mph wind (gust-prone), limited operations window

NOTE: All 7 days show 0.00" precipitation. Wind is primary constraint mid-week.

=====

FIELD WARNINGS

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— IONOSPHERIC DEGRADATION: TEC anomaly +2.2 TECU and data quality flag non-zero (Friday). Extend static occupation times by 10–15% or use post-processing with ionospheric corrections. Contact base office if cycle-slip rates exceed 2% per session.

— HEAT STRESS: High temperatures 95–106°F across entire week. Ensure crew hydration, shade rotation, and equipment thermal management (receivers, batteries). Peak intensity Tue–Wed.

— SATELLITE PRN 11: Currently unhealthy (UNUSUFN). Receivers will auto-exclude; no manual action required. Geometry (PDOP) already reflects this reduction.

— WEEKEND OPERATIONS: Saturday only viable weekend day (clear, calm). Sunday clouds add processing overhead. Crew availability/overtime authorization required if weekend work necessary.

=====

SITE CONDITIONS SUMMARY

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LAND COVER: Dwarf scrub, low sky obstruction, low multipath risk. Recommended elevation mask 13° (terrain-based). No wildfire activity within reporting range. All clear for field setup.

=====

OVERALL RECOMMENDATION

=====

DEPLOY FRIDAY AFTERNOON (3:00 PM start) if crew mobilization is complete; this is the optimal window for the entire analysis period. Primary week-long work schedule: Monday–Tuesday 3:00 PM–7:00 PM static windows with Friday as bonus high-priority session. Extend all sessions by 10–15% due to moderate ionospheric activity and Friday peak Kp 4.7. Hydration and thermal precautions mandatory given 95–106°F forecast. Thursday is backup only; avoid mid-week (Wed) due to elevated wind and overcast. Weekend reserve Saturday clear-sky window for makeup work if needed.

END OF REPORT

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Colusa County Clerk-Recorder

546 Jay St, Colusa, CA 95932

ASSESSOR / TAX OFFICE

Colusa County Assessor

546 Jay St, Colusa, CA 95932

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P270	HOPKINSLGHCN2005	3.0 mi
 NGS ↗	SUTB	SUTTER BUTTES	10.4 mi
 NGS ↗	CH01	CHICO 1	24.0 mi
 NGS ↗	CH05	CHICO 5	24.0 mi
 NGS ↗	CH02	CHICO 2	24.0 mi
 NGS ↗	CH06	CHICO 6	24.0 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	KT0441	"58 B"	18.660m / 61.2ft NAVD88	Order	0 mi
 Datasheet ↗	KT2287	"60 B RESET"	19.300m / 63.3ft NAVD88	Order	0.3 mi
 Datasheet ↗	KT0440	"W"	18.180m / 59.6ft NAVD88	Order	0.3 mi
 Datasheet ↗	KT0443	"X"	18.480m / 60.6ft NAVD88	Order	0.7 mi
 Datasheet ↗	KT1843	"GREEN"	15.400m / 50.5ft NAVD88	Order	1.9 mi
 Datasheet ↗	KT1844	"TROUGH"	15.810m / 51.9ft NAVD88	Order	2.3 mi
 Datasheet ↗	DL9151	"COLIND"	14.850m / 48.7ft NAVD88	Order	2.3 mi
 Datasheet ↗	KS1856	"COLUSA"	17.000m / 55.8ft NAVD88	Order	2.3 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: N/A

Effective: N/A

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
KT0441	58 B	39.215278	-122.013611	61.2	
KT2287	60 B RESET	39.213889	-122.008333	63.3	
KT0440	W	39.210361	-122.016250	59.6	
KT0443	X	39.212222	-122.001111	60.6	
KT1843	GREEN	39.188010	-122.023061	50.5	
KT1844	TROUGH	39.195512	-122.048830	51.9	
DL9151	COLIND	39.185125	-121.994591	48.7	
KS1856	COLUSA	39.192905	-121.981688	55.8	

[Download .txt File](#)

USGS LAKES & RESERVOIRS – within 150 miles

STATION	DIST	LEVEL	FULL POOL	LINK
CLEAR LK A LAKEPORT CA #11450000	49.7 mi	4.84 ft	N/R	USGS ↗

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Oroville	36.1 mi	855.92 ft (-44 ft)	900 ft	2,787K af	79%
Black Butte Lake	44.5 mi	—	228 ft	71K af	52%
Folsom Lake	57.5 mi	445.26 ft (-21 ft)	466 ft	758K af	78%
Shasta Lake	106.1 mi	1,021.24 ft (-46 ft)	1,067 ft	3,317K af	73%
Lake Del Valle	112.6 mi	696.47 ft (-10 ft)	706 ft	35K af	67%

USBR RESERVOIRS (Bureau of Reclamation) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Whiskeytown Lake	100.4 mi	1,209 ft*	1,209 ft	238K af	99%
Trinity Lake	114.6 mi	2,370 ft*	2,370 ft	1,943K af	79%
New Melones Lake	118.2 mi	1,088 ft*	1,088 ft	1,652K af	69%

ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Black Butte-Pool	47 mi	456.1 ft	458.36 ft	—	100%
Englebright Lake-Pool	52 mi	524.1 ft	522.35 ft	—	100%
Camanche	97 mi	230 ft	221.86 ft	—	104%
Del Valle	112 mi	696.4 ft	695.01 ft	—	100%
Farmington Dam-Pool	117 mi	124.6 ft	117.09 ft	—	106%
Boca	133 mi	5,601.2 ft	5,579.88 ft	—	100%

SNOTEL STATIONS (NRCS) – within 50 miles

STATION	DIST	ELEV	SWE	DEPTH	TEMP	ACCESS
Noel Spring [Snow Course]	41.4 mi	4,950 ft	0"	—	—	● Clear

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
014N001W28E001M	12.5 mi	364 ft	—	—	S100CNRLVL
014N001W30N002M	12.7 mi	394 ft	35.00 ft BGS	1960-02-23	S100CNRLVL
014N001W30N001M	12.9 mi	180 ft	44.00 ft BGS	1960-04-01	S100CNRLVL
014N001W32E002M	13.3 mi	508 ft	31.00 ft BGS	1966-10-14	S100CNRLVL
014N002W29J001M	13.3 mi	412 ft	—	—	S100CNRLVL
015N002E08R002M	13.8 mi	70 ft	35.00 ft BGS	1968-02-04	S100CNRLVL
014N002W34J001M	13.8 mi	463 ft	—	—	S100CNRLVL
014N001W32P001M	13.9 mi	100 ft	—	—	S100CNRLVL

The Central Valley aquifer system (S100CNRLVL) consists of Quaternary alluvial and fluvial sediments—sands, silts, and gravels—deposited by ancestral Sacramento River channels over the past 2 million years. This thick, layered sequence forms a highly productive groundwater reservoir because the coarser materials allow water to move freely while finer layers create confined zones that can sustain pressure and stable supplies. At your Colusa County location, water is readily available at moderate depths (30–50 ft in the 1960s), typical for the northern Sacramento Valley, though long-term regional pumping has caused gradual declines.

FORMATION

Name	Central Valley Quaternary Alluvial and Fluvial Deposits
Age / Rock	Quaternary (past 2.6 million years); unconsolidated alluvial sand, silt, gravel, and clay
Extent	est. 13,000–16,000 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	70–508 ft (7–51 floors)
Typical Depth	180–412 ft (18–41 floors)
Recharge Source	Precipitation and streamflow in Sierra Nevada foothills and local surface water; lateral flow from bedrock uplands
Transit Time	est. 20–50 years to 200 ft depth

WATER QUALITY

pH	est. 6.8–7.4 (est. USGS/DWR)
Hardness	est. 180–350 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 250–500 mg/L (est. USGS/DWR)
Notes	No sample data available; typical for formation—expect moderate hardness and neutral pH suitable for most uses

TREND

Trend	Water levels declined approximately 4–13 ft between February 1960 and October 1966, consistent with mid-20th-century regional drawdown from agricultural pumping
Source	USGS historical water-level records (1960–1968)
Seasonal Swing	est. 8–12 ft (est. USGS/DWR regional estimate)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓ **LOW SUBSIDENCE RISK — No known critically overdrafted basin**

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

🕒 **AI SUBSIDENCE ASSESSMENT — CLAUDE HAIKU**

The area at 39.2147, -122.0140 is located in the northern Sacramento Valley region with consolidated volcanic and sedimentary bedrock geology that has not experienced significant Quaternary subsidence. The Central Valley aquifer system (S100CNRLVL) in this locality is not critically overdrafted under SGMA, indicating groundwater extraction has remained within sustainable limits and has not induced permanent clay compaction. For surveyors operating in this area, historical benchmark elevations are expected to remain stable, and conventional survey methods are appropriate without subsidence corrections.

SUBSIDENCE ASSESSMENT

Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin

BENCHMARK IMPLICATIONS

Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate

SURVEY ACTION

Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	93 mph
Ground Snow Load	6.7 psf
Winter Wind (W2)	0.35
Ice Thickness	None
Frost Depth	6 in minimum
Brown's Formula · C=1 (Clay loam, Group C) · AFI=9°F-days · SF=1.25	
Verify with local building official before construction	

Source: ASCE 7-22 Hazard Tool · gis.asce.org

CLIMATE ZONE (IECC 2021): Zone 3B · Hot-Dry · Hot-Dry - warm/hot summers, cool winters, dry (Central CA, NV, AZ) County: Colusa

🕒 **AI CLIMATE SUMMARY — RESIDENTIAL DESIGN**

This hot-dry climate demands a tight, well-insulated envelope with minimal air leakage to resist extreme summer heat and low humidity, so prioritize shade, thermal mass, and strategic window placement—orient glazing toward north and east exposures while minimizing west and south-facing glass, or use deep overhangs and exterior shading if south glass is necessary for passive solar winter gain. Your key passive design move should be cross-ventilation through operable windows positioned to capture cool evening and night air, which is the most effective and affordable cooling strategy in this climate where nights cool significantly and humidity stays low year-round.

AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This site in central California presents moderate seismic risk with null hazard parameters suggesting either incomplete USGS data or a low-hazard location; wind exposure at 93 mph and modest snow load of 6.7 psf suggest standard structural design may be adequate. Moderately well-drained silt loam soils with shallow frost depth anticipated to support conventional spread footings, though a geotechnical investigation is recommended to confirm bearing capacity and confirm seismic site class.

FOUNDATION

Type Conventional spread footing anticipated

Basis Moderately well-drained silt loam (Hydro Group C) and shallow frost depth (6 in. planning estimate) suggest spread footing viability; consult a licensed engineer and geotechnical professional to confirm bearing capacity, settlement potential, and frost protection details

STRUCTURAL IMPLICATIONS

Seismic Seismic hazard parameters null - consult a licensed engineer and USGS hazard tools to confirm design basis earthquake parameters; Site Class D assumed for planning—geotechnical investigation recommended to refine

Wind Exposure Exposure C (open terrain) anticipated per ASCE 7-22 Risk Category II—verify local site conditions and terrain classification with design team

Wind 93 mph (3-sec gust) design wind speed anticipated—standard wind load design likely adequate; no hurricane or special wind hazard exposure anticipated

Snow Ground snow load of 6.7 psf anticipated—standard roof design likely adequate; consult engineer for any long-span or low-slope configurations

INVESTIGATIONS & INSPECTIONS

Geotech Required Yes—geotechnical investigation recommended to confirm Site Class D, bearing capacity, liquefaction screening, and frost/drainage conditions

Liquefaction Screen Screening recommended—consult engineer given Site Class D assumption and Hydro Group C drainage; shallow groundwater depth should be verified

Special Inspections Consult a licensed engineer to confirm whether special inspections are required based on final SDC determination and structural system selection

Code Edition ASCE 7-22 / IBC 2021 (planning reference only)

Site Class Note Site Class D assumed for planning—geotechnical investigation recommended to confirm soil profile, shear-wave velocity, and liquefaction potential

SOILS — USDA SOIL SURVEY

MAP UNIT: Moonbend silt loam, 0 to 2 percent slopes **SERIES:** Moonbend

DRAINAGE CLASS: Moderately well drained **HYDRO GROUP:** C

SOIL ORDER: Mollisols **SUBORDER:** Xerolls

AI SOIL INTELLIGENCE — CLAUDE HAIKU

Moonbend silt loam is a moderately well-drained Mollisol (Xeroll suborder) formed in alluvial and colluvial silt and clay deposits on nearly level valley floors and low terraces in semi-arid regions. This soil exhibits a dark surface layer typical of mollisols with subsurface clay accumulation, making it moderately reactive to moisture changes. Site visitors should anticipate seasonal water table fluctuations in winter/spring, stable but somewhat moisture-sensitive subgrades for construction, and the need for drainage design in areas with poor surface runoff control.

Alluvial and colluvial silt and clay parent materials on low terraces and valley floors under native grassland and shrub vegetation.

DRAINAGE

Class Moderately well drained

Seasonal Saturation Water table 30–48 inches in winter and spring; recedes below 48 inches in summer and fall

Flooding Risk Minimal on 0–2% slopes; site elevation and position limit flood risk

Ponding Risk None

ENGINEERING

Bearing Capacity 2,500–3,500 psf (shallow foundations); higher if subgrade moisture is controlled

Rating Fair

Shrink-Swell Moderate — silt loam surface with clay-rich substratum prone to minor volume change with seasonal moisture cycling

Frost Action Low to moderate — silt content and potential winter saturation warrant frost protection design in freeze-thaw zones

Cut Suitability Good — stable, cohesive material; minor slaking possible if exposed to wet-dry cycles

Fill Suitability Fair — clay substratum material requires moisture conditioning and compaction verification; not suitable as fill if saturated

SEPTIC

Suitability Conditional

Notes Perc test required — clay-rich substratum and seasonal water table (30–48 inches) may limit drain field performance; site-specific soil engineering and hydrogeologic assessment strongly recommended before septic design

HAZARDS

Excavation Low to moderate — silt and clay are generally stable when dry; high moisture content or seasonal saturation may cause sloughing; equipment access should account for potential softening in spring

Expansive Clay Low — clay content does not suggest high expansion potential, but moisture cycling may cause minor heave

Erosion Low on gentle slopes; localized gully erosion possible if surface runoff is concentrated during grading or construction

Other Seasonal water table elevation and spring runoff management are critical for long-term site stability and foundation performance

Data sources: USGS National Water Information System · waterdata.usgs.gov · California DWR / CDEC · cdec.water.ca.gov · USBR RISE · data.usbr.gov · USACE CWMS · water.usace.army.mil · NRCS SNOTEL/AWDB · wcc.sc.egov.usda.gov · USGS NWIS Groundwater · waterservices.usgs.gov · USDA Web Soil Survey · sdmdataaccess.nrcs.usda.gov · CA DWR SGMA Bulletin 118 · gis.water.ca.gov · ASCE 7-22 Hazard Tool · gis.asce.org · USGS Design Maps · earthquake.usgs.gov · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

Water levels current as of report run: Jul 17, 2026, 5:53 PM UTC

* Live pool elevation not reported by this reservoir — full pool design elevation shown

200 total **1** active **180** plugged By status: **180** Plugged **11** Canceled **8** Idle **1** Active

By type: 153 DH 40 DG 4 OG 3 GAS

Lease / Well	Status	Type	Operator	Depth	Dist
TXO Production Corp.	Plugged	DH	TXO Production Corp.	—	1.2 mi
Great Basins Petroleum Co.	Plugged	DH	Great Basins Petroleum Co.	—	1.3 mi
Venoco, LLC	Plugged	DH	Venoco, LLC	—	1.4 mi
Phillips Petroleum Company	Plugged	DH	Phillips Petroleum Company	—	1.7 mi
Whiting Oil and Gas Corporation	Plugged	DG	Whiting Oil and Gas Corporation	—	2.3 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	2.5 mi
Northern Michigan Exploration Company	Plugged	DH	Northern Michigan Exploration C...	—	2.6 mi
Union Oil Company of California	Plugged	DH	Union Oil Company of California	—	2.6 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	2.7 mi
Whiting Oil and Gas Corporation	Plugged	DG	Whiting Oil and Gas Corporation	—	2.7 mi
Western Continental Operating Company	Plugged	DG	Western Continental Operating ...	—	2.7 mi
Great Basins Petroleum Co.	Plugged	DH	Great Basins Petroleum Co.	—	2.7 mi
Natural Gas Corp. of Calif.	Plugged	DH	Natural Gas Corp. of Calif.	—	2.8 mi
Exxon Mobil Corporation	Plugged	DH	Exxon Mobil Corporation	—	2.8 mi
Natural Gas Corp. of Calif.	Plugged	DH	Natural Gas Corp. of Calif.	—	2.8 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	2.8 mi
California Resources Production Corporation	Idle	DG	California Resources Production...	—	3 mi
Richard S. Rheem, Operator	Plugged	DH	Richard S. Rheem, Operator	—	3 mi
Key Production Company, Inc.	Plugged	DG	Key Production Company, Inc.	—	3 mi
Whiting Oil and Gas Corporation	Plugged	DG	Whiting Oil and Gas Corporation	—	3.1 mi
Natural Gas Corp. of Calif.	Plugged	DH	Natural Gas Corp. of Calif.	—	3.1 mi
California Resources Production Corporation	Idle	DG	California Resources Production...	—	3.3 mi
Jacobson-Colusa Oil Co.	Plugged	DH	Jacobson-Colusa Oil Co.	—	3.3 mi
Amerada Hess Corp.	Plugged	DH	Amerada Hess Corp.	—	3.4 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	3.4 mi
Crest Oil & Gas Management Corp.	Plugged	DG	Crest Oil & Gas Management C...	—	3.5 mi
Crest Oil & Gas Management Corp.	Plugged	DG	Crest Oil & Gas Management C...	—	3.5 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	3.5 mi
Chevron U.S.A. Inc.	Plugged	DH	Chevron U.S.A. Inc.	—	3.6 mi
Pueblo Oil & Gas, Inc.	Plugged	DG	Pueblo Oil & Gas, Inc.	—	3.6 mi

Showing 30 of 200 wells. Full dataset via California.

AI Assessment — Well Proximity

Well proximity is not a primary planning concern for this site. The nearest well is a plugged TXO Production Corp. well 1.2 miles away, which falls outside the distance thresholds requiring foundation review or environmental flagging. With only one active/producing well in the 10-mile radius and no injection or disposal wells present, recommend a standard title search to verify whether any historical oil and gas easements or mineral rights encumbrances affect the parcel.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200
total in search area

1
within 5 mi

39
within 25 mi

Site Name	Status	Commodities	Distance
Cortina Mill	Producer	Sand/Gravel	0.6 mi
Lucky Boy Occurrences 2	Occurrence	Chromium	8.6 mi
Unnamed Prospects	Occurrence	Chromium	8.7 mi
Quarry	Unknown	STN	9.6 mi
Number Two	Unknown	Gold	10.1 mi
Secret Town	Unknown	Gold	10.2 mi
John Nugent Placer	Unknown	Gold	10.3 mi
San Juan Mine	Plant	Gold	10.4 mi
Myers Quarry	Producer	Stone/Crushed	10.7 mi
North Fork American River	Unknown	Gold	10.9 mi
Sutter Buttes Quarry	Producer	Sand/Gravel	11.4 mi
Sunset & Riverside	Occurrence	Gold	11.5 mi
Butte Ranch Borrow Site	Producer	Sand/Gravel	11.7 mi
Morehead Pit & Mill	Producer	Sand/Gravel	11.9 mi
Lower Sand Creek Bar	Producer	Sand/Gravel	12.7 mi
Pennsylvania	Unknown	Gold	12.9 mi
Butte Rock North Pit	Producer	Stone/Crushed	14.1 mi
Sand Creek Pit	Producer	Sand/Gravel	14.7 mi
Mcgilvray Quarry	Past Producer	Sand/Gravel	17.6 mi
Golden Gate Consolidated Qtz. Mine	Past Producer	Gold	17.7 mi
Antelope Valley Deposit	Occurrence	CLY_BN	18.6 mi
Salt Creek Bar	Producer	Sand/Gravel	19.1 mi
Robinson Construction Co	Producer	Sand/Gravel	19.6 mi
Mcfarland Pit and Mill	Producer	Sand/Gravel	19.8 mi
Gravel Pits	Unknown	Sand/Gravel	21.8 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

The project site has one active producer within 5 miles: Cortina Mill (0.6 mi), a sand and gravel operation. Plan for heavy truck traffic on access roads during operating hours and potential dust generation from this nearby aggregate operation. No blasting or crushing noise is expected unless the operation expands its processing capabilities.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

2
Endangered

2
Threatened

4
Total w/ Critical Habitat

2 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
Butte County meadowfoam	<i>Limnanthes floccosa</i> ssp. <i>californica</i>	Endangered

COMMON NAME	SCIENTIFIC NAME	STATUS
Vernal pool tadpole shrimp	<i>Lepidurus packardii</i>	Endangered
California tiger Salamander	<i>Ambystoma californiense</i>	Threatened
Yellow-billed Cuckoo	<i>Coccyzus americanus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Review **Location:** 39.2147, -122.0140 Finding ESA Section 7 consultation will be triggered if this project involves federal permits, funding, or approvals; the presence of four listed species within the 25-mile radius establishes a reasonable likelihood of adverse effects requiring formal biological assessment. **Butte County meadowfoam** and **vernal pool tadpole shrimp** warrant primary planning attention due to their narrow habitat specificity and endangered status, making them most vulnerable to site-level disturbance; **California tiger salamander** should also be considered given its potential use of seasonal wetland complexes common to this geographic area. The project should proceed with Phase 1 habitat assessment and coordination with USFWS to determine whether formal Section 7 consultation is required under the applicable federal nexus. Early environmental due diligence at the planning stage will clarify compliance pathways and avoid costly project delays. --- *For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.*

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

53	222	2026-07-07
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
alkali heliotrope	<i>Heliotropium curassavicum</i>	2025-08-15
alkali mallow	<i>Malvella leprosa</i>	2024-07-29
Alkali Weed	<i>Cressa truxillensis</i>	2024-05-15
big saltbush	<i>Atriplex lentiformis</i>	2024-01-16
Bonfire moss	<i>Funaria hygrometrica</i>	2023-11-30
box elder	<i>Acer negundo</i>	2025-05-21
broadleaf mistletoe	<i>Phoradendron macrophyllum</i>	2024-12-18
California brome	<i>Bromus carinatus</i>	2024-04-12
California mugwort	<i>Artemisia douglasiana</i>	2024-04-27
California wild grape	<i>Vitis californica</i>	2026-06-13
California Wild Rose	<i>Rosa californica</i>	2025-12-10
Common Madia	<i>Madia elegans</i>	2023-07-06
common tule	<i>Schoenoplectus acutus occidentalis</i>	2024-05-15
coyote brush	<i>Baccharis pilularis</i>	2023-09-06
cutleaf groundcherry	<i>Physalis angulata</i>	2025-07-15
Erect burhead	<i>Echinodorus berteroi</i>	2023-09-02
Frémont Cottonwood	<i>Populus fremontii</i>	2023-09-02
Giant Blazingstar	<i>Mentzelia laevicaulis</i>	2024-07-28
giant wild rye	<i>Leymus condensatus</i>	2026-03-02
heart-leaf milkweed	<i>Asclepias cordifolia</i>	2025-04-10
horseweed	<i>Erigeron canadensis</i>	2026-05-29
Miniature Lupine	<i>Lupinus bicolor</i>	2026-02-18
narrowleaf milkweed	<i>Asclepias fascicularis</i>	2023-09-02
Narrowleaf Willow	<i>Salix exigua</i>	2026-07-07
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2024-04-27
pale smartweed	<i>Persicaria lapathifolia</i>	2026-05-06
palmate salty bird's-beak	<i>Chloropyron palmatum</i>	2025-05-28
Purslane Speedwell	<i>Veronica peregrina</i>	2024-04-12
Red Ribbons	<i>Clarkia concinna</i>	2026-05-05
Red Ribbons	<i>Clarkia concinna concinna</i>	2023-06-14
redwhisker clammyweed	<i>Polanisia dodecandra</i>	2026-06-18
rough cocklebur	<i>Xanthium strumarium</i>	2023-09-02
Sacred Datura	<i>Datura wrightii</i>	2025-10-06
Saltgrass	<i>Distichlis spicata</i>	2025-07-03
sea clubrush	<i>Bolboschoenus maritimus</i>	2024-05-15
shield-bracted monkeyflower	<i>Erythranthe glaucescens</i>	2026-05-07
showy milkweed	<i>Asclepias speciosa</i>	2023-07-28
shy monkeyflower	<i>Erythranthe nasuta</i>	2025-04-11
Silvery Bryum	<i>Bryum argenteum</i>	2025-12-27

COMMON NAME	SCIENTIFIC NAME	LAST OBS
swamp smartweed	<i>Persicaria hydropiperoides</i>	2026-03-02
tall flatsedge	<i>Cyperus eragrostis</i>	2026-06-18
tarweed fiddleneck	<i>Amsinckia lycopsoides</i>	2024-04-12
Telegraphweed	<i>Heterotheca grandiflora</i>	2026-07-07
Toyon	<i>Heteromeles arbutifolia</i>	2026-01-22
turkey mullein	<i>Croton setiger</i>	2023-06-28
turkey tangle frogfruit	<i>Phyla nodiflora</i>	2025-07-04
twining snakelily	<i>Dichelostemma volubile</i>	2024-05-11
Upright Pepperweed	<i>Lepidium strictum</i>	2024-04-12
valley oak	<i>Quercus lobata</i>	2026-06-13
water fern	<i>Azolla filiculoides</i>	2025-07-03
Whiskerbrush	<i>Leptosiphon ciliatus</i>	2025-04-10
Wild Licorice	<i>Glycyrrhiza lepidota</i>	2024-06-06
woolly rosemallow	<i>Hibiscus lasiocarpus</i>	2025-07-15

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 39.2147, -122.0140 This site supports a **riparian and alkali wetland-transitional community**, dominated by moisture-tolerant species like Fremont cottonwood, common tule, and alkali-adapted plants (alkali heliotrope, big saltbush, alkali mallow). The ecology indicates a location near permanent or seasonal water sources with alkaline soils, likely in a valley or lowland setting with groundwater influence. Grading and drainage work should account for **seasonally high water tables and potential soil saturation**, particularly given the prevalence of hydrophytic species; inadequate drainage could exacerbate flooding or create anaerobic conditions, while excessive grading may disrupt the water regime supporting this specialized community. Soil pH testing and hydrological assessment are strongly recommended before any ground disturbance. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

25	30	9	82
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		25 obs	
Crotalus oreganus		25 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		30 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla aureola		9 obs	
Dasymutilla sackenii		9 obs	
TOXIC PLANTS			
SPECIES		OBSERVATIONS	
Toxicodendron diversilobum		39 obs	
Datura stramonium		21 obs	
Datura ferox		21 obs	
Datura wrightii		21 obs	
Conium maculatum		19 obs	
Urtica gracilis		3 obs	

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Briefing – Northern California Survey Area **Highest-priority hazards:** Northern Pacific rattlesnakes (**Crotalus oreganus**) present the greatest immediate risk in this region, followed by poison oak (**Toxicodendron diversilobum**), which is extremely common with 82 observations. Black widow spiders (**Latrodectus hesperus**) and various *Datura* species pose secondary chemical/envenomation risks. **Practical avoidance:** Wear closed-toe boots, long pants, and gaiters; watch hand and foot placement in brush and rocky areas where rattlesnakes shelter. Learn to identify poison oak (three leaflets, variable coloring) and avoid contact—it causes severe dermatitis. Do not touch or ingest any unfamiliar plants, especially flowering *Datura* species. Check gear, vehicles, and clothing regularly for spiders before entering enclosed spaces. **Seasonal note:** Late spring through early fall brings peak rattlesnake activity and poison oak allergenic potential; *Datura* blooms summer into fall. Maintain high vigilance May–October, though hazards persist year-round. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FAA AIRWAYS — ATS Routes crossing site area

22	8	6	8			
Total Airways	Jet Routes	Victor Airways	Other Routes			
AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	342.8°	161.8°
J126	Jet Route	High	18,000 ft	—	325.5°	144.0°
J143	Jet Route	High	18,000 ft	—	325.0°	145.8°
J189	Jet Route	High	18,000 ft	—	335.5°	155.0°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	22,000 ft	—	342.6°	162.6°
J65	Jet Route	High	18,000 ft	—	304.1°	121.1°
Q1	Q-Route	High	—	—	334.5°	154.1°
Q120	Q-Route	High	—	—	38.1°	219.8°
Q122	Q-Route	High	—	—	37.8°	219.6°
Q124	Q-Route	High	—	—	37.8°	219.6°
Q138	Q-Route	High	—	—	47.7°	229.0°
Q3	Q-Route	High	—	—	164.0°	344.6°
Q5	Q-Route	High	—	—	162.7°	343.3°
T259	T-Route	Low	—	—	340.5°	160.2°
V108	Victor Airway	Low	4,500 ft	—	117.7°	296.8°
V150	Victor Airway	Low	3,000 ft	—	35.4°	215.5°
V195	Victor Airway	Low	5,300 ft	—	352.9°	172.9°
V199	Victor Airway	Low	9,000 ft	—	21.3°	199.6°
V200	Victor Airway	Low	6,200 ft	—	72.5°	252.8°
V23	Victor Airway	Low	2,000 ft	—	293.8°	113.6°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

3						
Rail Lines						
OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION	
CFNR	Mainline	1	No	No	WEST VALLEY	
CFNR	Other	1	No	No	—	
CFNR	S	1	No	No	—	

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200	87	69	8	8	2	
Total Observations	Total Species	Birds	Mammals	Reptiles	Amphibians	
BIRDS (69 species)						
COMMON NAME	Scientific Name					OBS
Great Egret	Ardea alba					8
Black-crowned Night Heron	Nycticorax nycticorax					8
Western Cattle-Egret	Ardea ibis					7
Red-winged Blackbird	Agelaius phoeniceus					6
Killdeer	Charadrius vociferus					5
Bald Eagle	Haliaeetus leucocephalus					5
Mourning Dove	Zenaida macroura					4
Wild Turkey	Meleagris gallopavo					4
House Sparrow	Passer domesticus					4
Bullock's Oriole	Icterus bullockii					4
House Finch	Haemorrhous mexicanus					4

COMMON NAME	Scientific Name	OBS
Great-tailed Grackle	<i>Quiscalus mexicanus</i>	3
Red-tailed Hawk	<i>Buteo jamaicensis</i>	3
Western Kingbird	<i>Tyrannus verticalis</i>	3
Pied-billed Grebe	<i>Podilymbus podiceps</i>	3
+ 54 additional species		

MAMMALS (8 species)

COMMON NAME	Scientific Name	OBS
Columbian Black-tailed Deer	<i>Odocoileus hemionus columbianus</i>	4
Black-tailed Jackrabbit	<i>Lepus californicus</i>	4
North American River Otter	<i>Lontra canadensis</i>	3
Mule Deer	<i>Odocoileus hemionus</i>	3
Coyote	<i>Canis latrans</i>	1
Pronghorn	<i>Antilocapra americana</i>	1
American Badger	<i>Taxidea taxus</i>	1
Domestic Cattle	<i>Bos taurus</i>	1

REPTILES (8 species)

COMMON NAME	Scientific Name	OBS
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	5
Western Fence Lizard	<i>Sceloporus occidentalis</i>	3
Mountain Garter Snake	<i>Thamnophis elegans elegans</i>	3
Gopher Snake	<i>Pituophis catenifer</i>	2
Northwestern Pond Turtle	<i>Actinemys marmorata</i>	2
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	2
Red-eared Slider	<i>Trachemys scripta elegans</i>	1
Northern Pacific Rattlesnake	<i>Crotalus oreganus oreganus</i>	1

AMPHIBIANS (2 species)

COMMON NAME	Scientific Name	OBS
Western Toad	<i>Anaxyrus boreas</i>	2
Western Spadefoot	<i>Spea hammondi</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Assessment Summary This site in northern California (Clear Lake vicinity) supports a diverse wildlife community characteristic of mixed grassland, shrubland, and wetland/riparian habitats with moderate to high disturbance tolerance. The abundant wading birds (egrets, herons) and presence of aquatic species (river otter, pond turtle, spadefoot) indicate proximity to permanent or seasonal water bodies and wetland complexes, while terrestrial species (pronghorn, badger, gopher snake) reflect open grassland and shrub-steppe conditions. Notable regulatory considerations include the Bald Eagle (federally protected), pronghorn (California Department of Fish and Wildlife Species of Special Concern), and Western Spadefoot (sensitive amphibian dependent on vernal pools or seasonal wetlands), all of which warrant habitat protection and avoidance during project activities. The overall species composition suggests a site with intact grassland-wetland mosaic habitat supporting native species, though the presence of common generalists and non-native species (red-eared slider, house sparrow, domestic cattle) indicates existing land use pressures and moderate anthropogenic disturbance compatible with ranching, agriculture, or rural development. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳️ PASSIVE + ACTIVE SOLAR PLANNING — Lat 39.21°

1 • AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.6 hrs	Overall solar resource
Summer average	9 hrs	June–August
Winter average	3.9 hrs	December–February
Worst month (Dec)	3.2 hrs	Size battery storage to this

2 • IDEAL SOLAR ARRAY PITCH

Optimal tilt = 39.2° (latitude) → 10/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	96.9%	good
5/12	22.6°	97.5%	good
6/12	26.6°	98.1%	→ excellent
7/12	30.3°	98.7%	→ excellent
8/12	33.7°	99.2%	→ excellent
9/12	36.9°	99.7%	→ excellent
10/12	39.8°	99.9%	→ annual optimal
12/12	45°	99.1%	→ excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +723 kWh/yr (~\$108 value/yr). Over 25yr panel life: ~\$2710 total gain.

3 · SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	49	30	Minimize south glass
South glass (SHGC 0.6)	73	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	26	104	△ Minimize — afternoon load
Roof — dark (absorb 0.95)	233	301	Highest load surface
Roof — light (absorb 0.35)	86	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	61	79	Best roof option for heat
South wall — dark	91	69	
South wall — light	30	23	Light color = major savings
Hardscape — asphalt	181	301	△ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.
Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 · ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 · THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3B Hot-Dry: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 · TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.2	OK
20ft	All year	2.2	2	2.4	OK
30ft	All year	3.1	2.9	3.8	Zone 2 check
40ft	All year	4.1	3.8	5.4	△ verify
50ft	All year	4.9	4.7	7.8	△ verify
60ft	All year	5.7	5.5	9.2	✗ Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†Balanced = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🕒 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan You're in a fantastic location for solar—with about 6.6 peak sun hours every day on average, your panels will generate solid power year-round, especially in summer when you hit 9 hours. Think of peak sun hours like the number of hours your panels work at full strength; it's much better than many parts of the country. Now, your roof color is surprisingly important: a dark roof absorbs intense summer heat that then radiates into your home, but painting it light or installing light-colored panels can cut that unwanted heat load by two-thirds. Your roof's pitch of about 39 degrees is nearly perfect for catching the sun, so you're lucky there. Here's a practical warning: your west-facing windows are your home's weak spot in the afternoon, because western sun pumps almost twice the summer heat through them compared to south-facing glass, so you'll want shading (a pergola, exterior shade screen, or even strategically-placed plants) on that side. A deciduous tree planted on your south side is genuinely smart because it'll shade your roof and windows in scorching summers (blocking about 4 hours of direct heat) while dropping its leaves in winter to let warming sun through—*though if you're in a wildfire zone, check your local fire code first, since some areas restrict trees near homes*. Thermal mass is simply material that soaks up heat during the day and releases it slowly at night, which keeps your home more comfortable without constantly running air conditioning. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING 39.21471°N, 122.01399°W | Colusa County, CA

Structural Design Drivers & Climate Envelope

This site falls within IECC Climate Zone 3B (Hot-Dry) with an annual freezing index of 9°F-days, placing it at the minimal frost threshold. Wind, snow, and seismic design parameters are not available from the provided dataset, limiting immediate determination of the controlling load case. In the absence of those values, the hot-dry climate classification and minimal freeze-thaw cycle frequency suggest thermal and desiccation effects may warrant closer attention than in colder zones, but the structural system selection cannot be finalized without wind speed, seismic acceleration, and snow load data specific to this coordinate. Request USGS seismic hazard maps and ASCE 7 wind speed contours for the Colusa County location to establish baseline lateral and vertical design loads.

Geotechnical Foundation Requirements

The site is underlain by Moonbend silt loam with 0–2 percent slopes, classified in USDA Hydrologic Group C with moderately well-drained status. Hydro Group C soils typically exhibit moderate to slow infiltration rates; the moderately well-drained classification indicates acceptable percolation but requires verification of seasonal groundwater behavior. Frost depth of 6 inches—derived from Brown's Formula using the 9°F-day AFI—establishes minimum footing depth at 6 inches below grade for frost protection. However, silt loam in Group C may be susceptible to frost heave if moisture migration occurs during the minimal freeze cycles; subsurface drainage design should prevent water accumulation above the footing bearing plane. Bearing capacity and settlement must be confirmed through site-specific subsurface exploration, as texture data for the Moonbend series are incomplete in this briefing.

Hazard Assessment

No Quaternary faults are present within the search radius, and no subsidence basin data are recorded for the site vicinity. Seismic hazard cannot be evaluated without acceleration or hazard level data. One active oil/gas well (TXO Production Corp.) is located 1.2 miles away with a plugged downhole status; the producing state of this well and any induced seismicity risk should be confirmed with California Department of Conservation records. Mining activity is significant regionally—200 MRDS deposits exist within approximately 35 miles—but only one active producer, Cortina Mill, operates within 5 miles at 0.6 miles from the site. Ground stability impacts from mining subsidence or vibration are unlikely at this distance but should be verified against Cortina Mill's extraction methods and depth. Wildfire hazard data were not provided; request California Fire and Resource Assessment Program (FRAP) mapping for the site.

Site Context & Preliminary Risk Profile

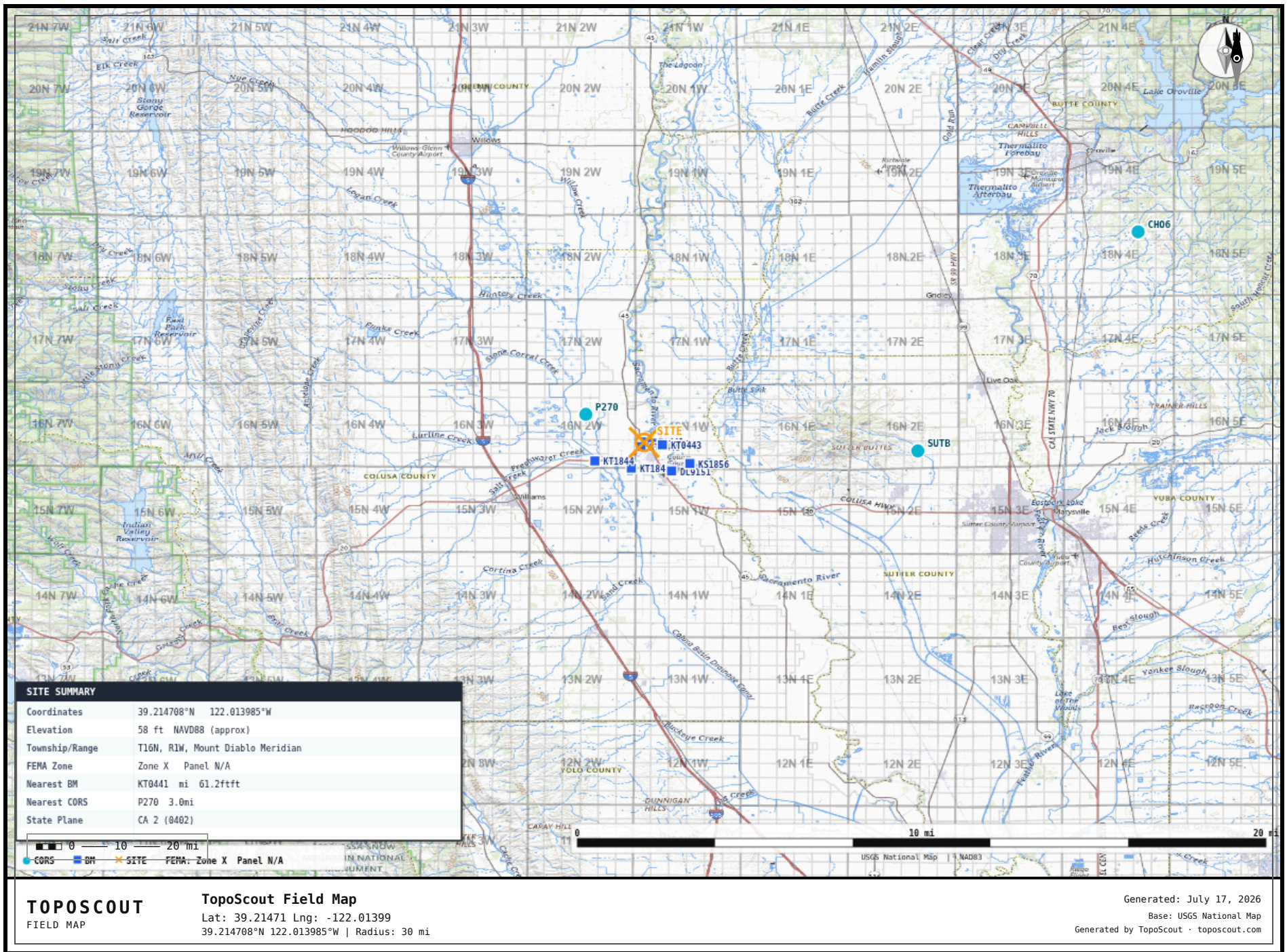
Colusa County soils in this zone are agricultural-grade, and the minimal frost depth reflects low winter severity typical of California's northern valley margin. The combination of Hydro Group C drainage and hot-dry climate suggests good long-term drainage but requires confirmation that seasonal perched water does not develop above the silt loam bearing layer. The proximity of industrial operations (mining 0.6 mi, plugged oil well 1.2 mi) is noted but does not present imminent structural risk on current data. The site's position in a seismically active state demands complete seismic hazard characterization before final design.

Bottom Line: Obtain USGS seismic hazard acceleration (MCE and Design) and ASCE 7 wind speed parameters for this coordinate, confirm subsurface stratification and seasonal groundwater elevation via Phase I geotechnical boring, and verify Cortina Mill's operational status and extraction depth relative to site stability.

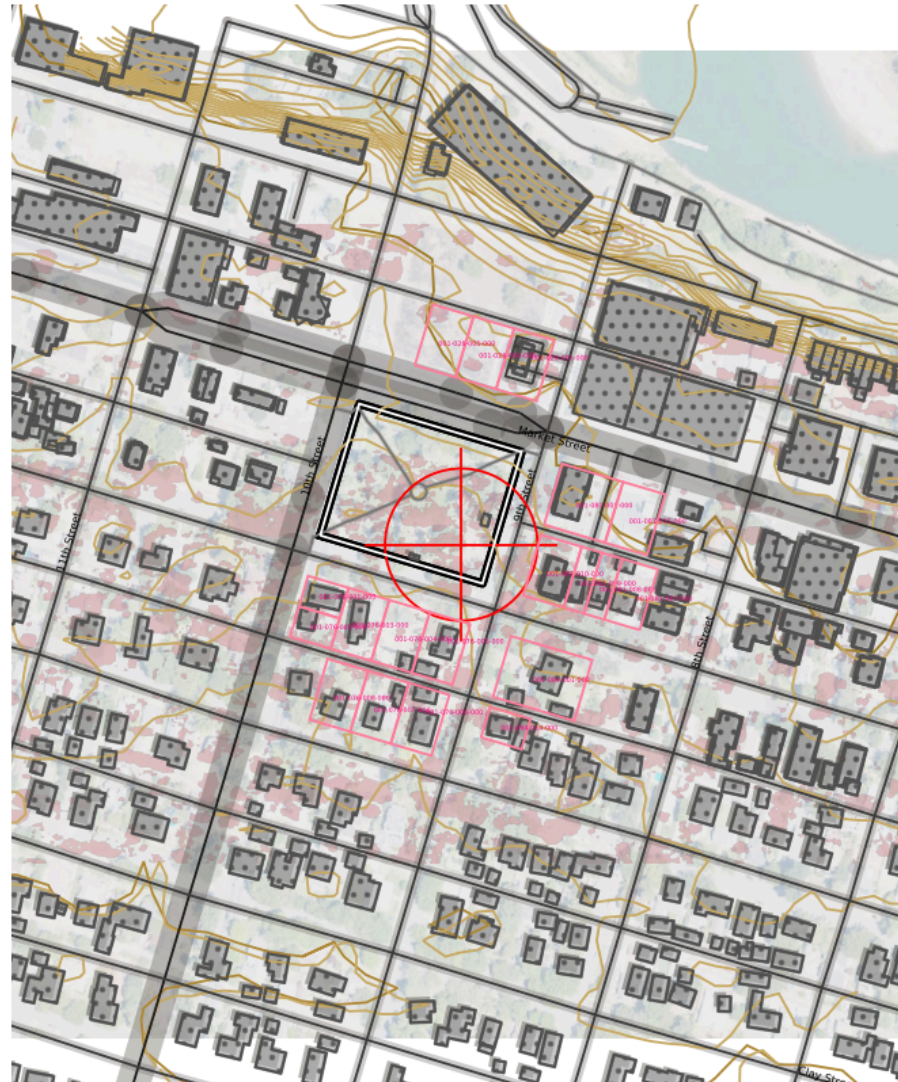


Copy to clipboard

Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.



TOPOSCOUT SITE PLAN PREVIEW



INCLUDED LAYERS

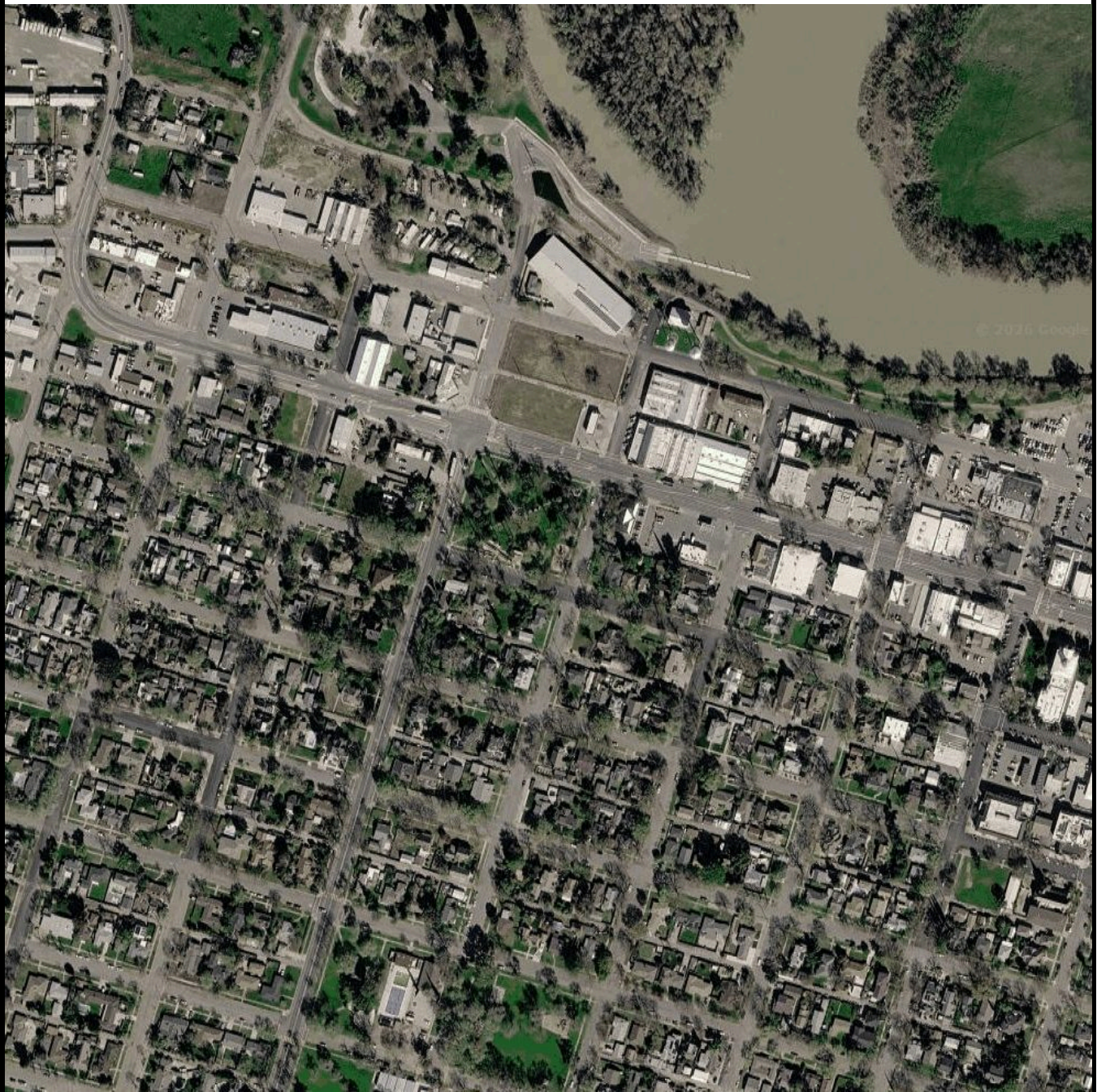
- Contours (Index/Major/Minor/1ft)
- PLSS Section/Township/QQ
- Roads Major/Local/Trail + ROW
- Buildings
- Hydro
- Power Lines
- Pipeline
- Railroad
- Benchmarks + Labels
- County Boundary
- OTLS Grid
- Site Info
- Elevation Labels
- 3D Polylines (all geometry)

N
0 — 400 ft State Plane · NAD83 · US Feet



SITE AERIAL – 600 ft × 600 ft

39.21471°N, 122.01399°W · Google Aerial



AREA CONTEXT — 3000 ft × 3000 ft

39.21471°N, 122.01399°W · Google Aerial

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

